

MARS HACKATHON

Mission Brief

Tuesday 15 September · 17:30 – 22:00 · London
PhysicsX · GirlsWhoML · supported by Cursor

01 The Premise

Commercial spaceflights reach Mars, and for the first time a civilisation gets to be *planned* rather than inherited. No sprawl to undo, no grid laid down by accident, no century of mistakes to clean up. A blank planet — and a rare chance to get it right on purpose.

The people who plan a trip this far ahead are long past survival mode. The real question is harder: how do you build a place people actually want to *live* — one that supports both livelihood and leisure, work and rest, systems and streets — without repeating the reflex that shaped Earth: taking whatever is nearest, as fast as possible, and paying for it later? So, is a luxury hotel really the first thing Mars needs — or the last?

You have one evening to make your case. Pick one track and design a working piece of a Martian city that treats the planet as something to build *with*, not through. Use whatever combination of engineering, ML, or 3D modelling gets you there. Everyone gets **\$50 in Cursor credits**, mentors are on the floor throughout, and a panel of industry judges reviews what you build. Come as a team of 2–3, or show up solo and join one at team formation.

02 Pick Your Track

Commit to one track. Each is a different layer of a working city — the shelter people live in, the way they move, and the systems that quietly sustain both. Use machine learning to make one real decision better than a guess.

Architecture — habitats and shelter

Everything that makes a place genuinely liveable inside a hostile environment: pressurised living space, radiation shielding, insulation against -60°C nights, and structures printed or sintered from the regolith underfoot rather than hauled 60 million km from Earth.

Where ML helps. Optimise wall thickness against radiation and internal pressure, sequence a modular pod layout for the fastest liveable volume, or predict the thermal stability of a lava-tube shelter across the day–night cycle.

Example builds. A printed regolith dome; a modular pod colony; a sheltered sub-surface habitat.

Vehicles & Mobility — moving across Mars

Getting people, cargo, and robots across broken, unmapped ground on strictly limited power. Rovers, transporters, and autonomous fleets that have to choose safe routes and spend every watt carefully.

Where ML helps. Classify terrain from imagery, plan an autonomous route scored by safety and energy cost, or balance payload against battery range over rough ground.

Example builds. A route-scoring rover; a cargo pod that trades load for range; a swarm that digs, hauls, and builds in parallel.

Life Support & Resource Systems — the closed loop

The systems that make a settlement self-sufficient: energy, water, breathable oxygen, and food. On Mars these must run as tight closed loops, turning buried ice and a thin CO₂ atmosphere into resources instead of hazards.

Where ML helps. Forecast ice yield by location, run a microgrid that predicts demand and allocates power, or tune light, water, and CO₂ in a closed greenhouse.

Example builds. An ice-yield map; a demand-aware solar microgrid; a self-balancing greenhouse.

03 Tools & Resources

A starting kit of open-source libraries and real Mars data. You do not need all of it — pick the one or two that fit your track. Everything below is free, and most runs from Python or the browser.

Mars environment, terrain & data

Tool	What it gives you
SpiceyPy (NASA SPICE)	Planetary geometry, positions, time and coordinate frames — where Mars, the Sun and your site are, and when.
GDAL / rasterio	Read real Mars elevation and imagery (HiRISE DTMs, MOLA) as terrain rasters you can slice and analyse.
OpenPlanetary / Mars Trek	Open Mars basemaps, GIS layers and place data for siting a base or building a map.
NASA Planetary Data System	The raw mission archive — imagery, spectra and terrain from real Mars missions.
pvlb	Model solar power output; adaptable to Martian irradiance for sizing panels and a microgrid.

Simulation, physics & robotics

Tool	What it gives you
Gazebo + ROS 2 (Nav2)	Simulate a rover on rough terrain with realistic physics; plan and test autonomous navigation.
PyBullet / MuJoCo	Lightweight physics sandboxes for wheeled robots, arms and rigid-body dynamics — quick to prototype.
OpenFOAM	Computational fluid dynamics for airflow, dust and thermal behaviour inside a habitat or greenhouse.
FEniCS / CalculiX	Finite-element structural analysis — stress and deflection in a printed wall or pressurised shell.

CAD & 3D modelling

Tool	What it gives you
Blender (bpy)	Free modelling, visualisation and rendering; scriptable in Python. Best for habitat concepts and city-scale scenes.
FreeCAD	Full open-source parametric CAD for engineering-accurate parts with real dimensions and tolerances.
CadQuery / build123d	Define precise CAD models in Python code — ideal if you would rather script parts than click them.

Tool	What it gives you
OpenSCAD	Script-based parametric CAD for clean, repeatable geometry.
Onshape	Browser CAD, nothing to install, free for public projects. A sensible default for the Vehicles and Life Support tracks.

Machine learning

Tool	What it gives you
scikit-learn	Fast, reliable classification, regression and forecasting — the quickest path to a working model in a few hours.
XGBoost / LightGBM	Strong tabular forecasting for ice yield, power demand or terrain scoring.
PyTorch / TensorFlow	Deep learning when you need it — image-based terrain classification, for example.
Optuna	Search a design space automatically — tune wall thickness, route weights or greenhouse settings.

04 Mission Schedule

Time	Phase	Detail
17:30	Launch pad	Doors open, food, crew formation — show up solo and find a team here.
18:00	Mission briefing	PhysicsX intro (10 min), GirlsWhoML & challenges (10 min), Q&A, judging rubric published.
18:20	Ignition	Hacking begins — a 2 h 10 m window, mentors on the floor.
20:00	Mid-mission check	Optional 5-min mentor check-in per crew, non-disruptive.
20:30	Code freeze	Push to shared GitHub — hard cutoff 20:35.
20:38	Demos	3-min build plus 2-min Q&A per crew.
21:30	Judging	Room judges deliberate; top 3 selected per room.
21:45	Close	Winners announced, prizes, wrap — venue clear by 22:00.

05 Suggested Checkpoints

Self-check goals layered onto the schedule, not extra events. The 20:00 target matters most: get something functional before you polish. Running out of time with nothing that works is the failure mode of every short hack.

By	Checkpoint	Crew should have
18:35	Scope locked	Chosen category, one clear build, and the single decision ML will drive. Repo created, README started.
19:15	Skeleton up	A rough 3D model or diagram, plus a data plan: what the ML predicts or optimises, and what feeds it.
20:00	Core working	The ML component runs end to end, even if crude, producing a result that informs the design.
20:25	Freeze-ready	Demo path decided, README updated with category and ML usage, final commits staged before 20:35.

By	Checkpoint	Crew should have
20:38	Demo	3-min walkthrough plus 2-min Q&A.

06 Rules of Engagement

Crew size. 2 to 3 people. Solo is allowed. No crews of 4 or more.

Credits. Everyone gets \$50 in Cursor credits to build with.

Submission. Push to a shared GitHub repository — hard cutoff at 20:35, enforced by commit timestamp.

Theme. Building toward a Martian city, with the option to use 3D modelling and ML.

Mentors. On the floor 18:15–20:30, covering PhysicsX expertise plus 3D-modelling and ML support.

Judging. Rubric posted at 18:00 so it is known before coding starts; 3–5 judges in-room 20:38–22:00.

Prizes. From PhysicsX, including career coaching and mentorship, with more to be announced.

07 Judging Criteria

Rubric published at 18:00, before coding begins.

#	Criterion	What we're looking for	Score
1	Impact & Purpose	Does the build solve a real problem the first settlement would actually face? Is it something the city needs before comfort or novelty?	/10
2	Innovation & Creativity	A fresh take on surviving Mars? Does it use local resources, ML, or design in a clever, unexpected way?	/10
3	Technical Execution	How well is it built in the time given? Does the model, code, or CAD actually run, and does the demo hold together?	/10
4	Feasibility & Real-World Potential	Could this credibly scale from a first landing toward a real settlement? Is the path from prototype to deployment believable?	/10
5	Team & Collaboration	Effective use of diverse skills? Genuine cross-disciplinary work between technical and non-technical builders. (Bonus /5 for pitch.)	/10
	TOTAL		/50

08 GitHub Submission

On the night, each crew creates a public repo under the event org, named like **mars-hack-crew-name**, and commits code, models and design files as they go. Include a short **README.md** covering what you built, which category, how ML was used, and how to run or view the demo. The hard cutoff at 20:35 is enforced by the last commit timestamp — anything pushed after that is not judged. A one-line setup guide and a ready-made template repo link are shared at 18:00 to remove friction for less Git-confident crews.